

Leetcode :

Functional requirements:

- user should be able participate in the coding contest,
- user should be able to submit solution for any problem based on his preference,
- user should be able to search for any problem based on his preference,

Non-functional requirements:

- data-partitioning,
- load balancing,
- fault tolerance,
- vertical scalability,
- horizontal scalability,
- low latency,
- high throughput,
- CAP theorem (Here we need to prioritise Strong Consistency for rank predictions during Coding Contest, & for search scenario we need to prioritise 99.99% uptime i.e Availability>>Consistency)

Core Entities:

- user,
- problem,
- history,
- contest

APIs:

POST:/submit/solution
{
 userID,
 {fileMetadata}
}
GET:/search/{problemID}

HLD :

